

PRABHAKARA RAJU MADDI

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Objective

Computer Science undergraduate with strong foundations in Machine Learning, Data Science, and Algorithms. Experienced in predictive modeling, NLP, Retrieval-Augmented Generation (RAG), and statistical data analysis. Passionate about building intelligent systems, applying machine learning to real-world problems, and conducting data-driven experimentation.

Education

Anil Neerukonda Institute of Technology and Sciences
B.Tech in Computer Science and Engineering

Sep 2023 – Present
CGPA: 9.63/10

Experience

AI Azure Intern – Edunet Foundation Microsoft Azure

May 2025

- Built and evaluated supervised machine learning models using structured datasets.
- Performed data preprocessing, exploratory data analysis, and feature engineering.
- Worked with Azure AI services for experimentation and deployment workflows.
- Documented evaluation metrics, findings, and technical observations for model assessment.

Projects

AI Chatbot (RAG) *LangChain, Hugging Face, NLP*

- Developed a Retrieval-Augmented Generation framework for knowledge-grounded question answering.
- Evaluated embedding models and vector similarity search techniques for semantic document retrieval.
- Implemented chunking, retrieval, and reranking strategies to improve contextual relevance.
- Analyzed retrieval performance and response quality through systematic experimentation.

Renewable Energy Forecasting *Python, Scikit-learn, Pandas, NumPy*

- Built machine learning models for renewable energy output prediction using weather and historical generation datasets.
- Performed feature engineering, missing value treatment, and correlation analysis.
- Evaluated Linear Regression, Random Forest, and Gradient Boosting models using RMSE and MAE metrics.
- Conducted comparative analysis of forecasting approaches and identified key influencing factors.

Customer Churn Prediction *Python, Scikit-learn, Pandas, NumPy*

- Developed a supervised machine learning pipeline to predict customer churn.
- Performed data cleaning, feature engineering, exploratory data analysis, and class imbalance handling.
- Trained and compared Logistic Regression, Random Forest, and XGBoost models.
- Evaluated models using Accuracy, Precision, Recall, F1-score, and ROC-AUC metrics.

Telugu Speech Chatbot *Python, NLP, Speech Processing*

- Built a multilingual speech-to-speech conversational system focused on Telugu interactions.
- Integrated Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) pipelines.
- Designed NLP-based intent recognition workflows for conversational understanding.
- Explored low-latency speech processing techniques for real-time interactions.

Technical Skills

Languages: Python, Java, C, C++, SQL

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Time-Series Forecasting

AI/NLP: Retrieval-Augmented Generation (RAG), NLP, Prompt Engineering, Embeddings

Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face, LangChain

Databases: MongoDB, SQLite

Tools: Git, GitHub, Azure AI Services, Jupyter Notebook, Postman

Certifications

- Oracle Certified AI Foundations Associate (2025)
- Data Science for Engineers – NPTEL (2026)
- AI Fluency Certification – Anthropic (2026)